

The “Problem of Universals”

March 18th, 2004

1 Explaining resemblance

In the previous class, we ended up characterising the ‘problem of universals’ as a *challenge* to answer certain ‘what is it to be...?’ questions, such as the following:

- (1) What is it for two things to be of the same type?
- (2) What is it for two things to resemble one another?
- (3) What is it for two things to resemble one another *in some respect*?
- (4) What is it for two things to resemble one another *perfectly*, or to be duplicates?

Of these challenges, I think (4) is the sharpest and best-focused. As regards (1), there’s the problem that a nominalist might reply: ‘Strictly speaking, there are no types, so it never happens that two things are of the same type, so the question doesn’t arise or can be given a trivial answer’. (A similar point can be made about the talk of ‘respects’ in (3).) (1), (2) and perhaps (3) are also manifestly *vague* and *context-dependent*, and it’s not clear how we answer ‘what is it to be...?’ questions involving vague predicates. (3) can, arguably, be answered ‘Since any two things resemble one another in some respect, the question only needs a trivial answer.’ But (4) doesn’t seem to be problematic in any of these ways.

2 Real definitions

Questions of the form ‘What is it for something to Φ ?’, or ‘What is it for one thing to Ψ another thing?’ are requests for what are sometimes called (in a usage that dates back to Locke) *real definitions*.

Real definitions don’t have to be knowable a priori. ‘To be made of water is to be composed of H₂O molecules’ is a perfectly good, but *a posteriori*, real definition.

Discussions about real definition are often conducted in a vocabulary that takes the existence of properties for granted. Instead of ‘To be made of water is to be composed of H₂O molecules’, one says ‘The property of being made of water is the property of being composed of H₂O molecules’.

This is all very well in many contexts, but it is apt to be misleading in the present one. It may misleadingly suggest that a nominalist who doesn’t think there are any properties (strictly speaking) doesn’t have to be concerned with questions of real definition.

Warning: Armstrong often talks in these terms!

There seems to be something faulty about *circular* real definitions. If I want to say that a line is the shortest distance between two points, I can't also say that to be a point is to be the intersection of two lines! Similarly, there seems to be something faulty about an *infinite regress* of real definitions (examples of this are a bit obscure: we'll come across one in due course when we talk about "Bradley's regress").

It follows from this that *definitions must eventually come to an end*: as one keeps asking questions of the form 'What is it to be F ?', one will eventually reach the point at which there is no longer anything to say: one can't explain in simpler terms what it is to be F , because there is nothing 'more basic' in terms of which one could give one's explanation. When this happens, we say that the predicate ' F ' is *primitive*.

One very useful way to classify different kinds of Nominalist and Realist is to ask *which predicates they take as primitive*.

If you look at what someone says in answer to the challenge to explain what it is for two things to resemble one another, you'll often get a good sense what predicates that philosopher takes as primitive.

3 Armstrong's shift

Armstrong proceeds to talk about the view he calls 'Class Nominalism'. But, confusingly, this view doesn't seem to take the form of an answer to any of these questions. Here are some things Armstrong says in characterising the view of the Class Nominalist:

for a thing to be of a certain type is nothing more than for it to be a member of a certain class

According to Class Nominalism, for a to be an electron is nothing more than for it to be a member of the class of electrons

Now Armstrong seems to be talking about some rather different challenges: the challenge to answer such questions as these

(5) What is it for something to be an electron?

(6) Whenever t is a type, what is it to be of t ?

(6) strikes me as a pretty dubious. For one thing, since it quantifies over types, it will be open to the Nominalist to reply: ‘I reject the question on the grounds that there are no such things as types.’ But more importantly, it carries the presupposition that there must be some *general schema* we can use to answer *any* question of the form

(7) What is it to be F ?

at least in those cases where the predicate ‘ F ’, as we would ordinarily put it, ‘picks out a type’.

It’s not at all clear that anyone—and especially the Nominalist—should endorse such a general schema. It seems quite reasonable to me to say that the question ‘What is it to be an F ?’ has to be answered in a case-by-case way. It’s hard to see what sort of general schema could generate such apparently true claims as the following:

To be a square is to be a quadrilateral with equal sides and angles.

To be a hydrogen atom is to be something composed of a single proton and zero or more electrons orbiting that proton.

Of course, these claims don’t have to be the end of the story: we can go on to ask what it is to be a quadrilateral, or what it is to be a proton, and use the answers to come up with even ‘deeper’ analyses of what it is to be a square or to be an electron. But to reject the demand for a general schema, we only need to think that this could at least be the *beginning* of the story.

By contrast, (5) ('What is it to be an electron?'), seems to me to be a perfectly good and important question.

One might wonder why *philosophers* should be interested in this question: isn't it a question for the scientists?

Well, it's true that science *might* yield some analysis of 'electron' in terms of other predicates taken from particle physics: e.g. 'to be an electron is to be a "string" vibrating in such-and-such pattern'. But then the 'what is it to be...?' question will arise again for whatever vocabulary we used in this scientific definition.

The real philosophical question here might be expressed like this: 'What method or strategy should we use to answer those questions of the form "What is it to be *F*?" that remain when all of the standard scientific methods have been exhausted?'

But we can harmlessly go along with the pretence that 'electron' is fundamental *as far as science is concerned*: the standard methods of science aren't going to yield any interesting answer to the question 'What is it to be an electron?'

4 Class Nominalism: some points

Let's go back to the formula

(8) To be an electron is to be a member of the class of electrons

Isn't this analysis circular? No: what's really intended is this:

(9) The class of electrons is such that to be an electron is to be a member of *it*.

As Armstrong sets things up, the Class Nominalist is really committed to the existence of *classes* (sets). But there is a closely related position that can be adopted even by the Nominalist in the first sense, who doesn't believe that there are (strictly speaking) any such things as sets. One can use *plurals* to do the work of sets, saying

(10) The electrons are some things such that to be an electron is to be one of *them*.

Alternatively, one can use an *infinite disjunction*, saying

(11) To be an electron is to be identical to e_1 , or e_2 , or ... or e_n .

Of course we could never actually *state* this definition since we don't have names for all the electrons there are, but we can still state that 'electron' has a real analysis *of this form*.

5 Armstrong's objection to Class Nominalism

Armstrong objects that 'there is no reason to think that to every class there corresponds a type'. But it's not clear that the Class Nominalist has to be committed to saying this.

One way to think of the point Armstrong is making here is to say that the Class Nominalist has no good way to answer questions like

- (1) What is it for two things to be of the same type?

We can't say

- (12) For x and y to be of the same type is for there to be some class that has both x and y as members.

since that entails, in conjunction with the axioms of set theory, that any two things are of the same type.

Another way to put the point: there are two different kinds of classes, those that 'mark off a type' from those that don't. Armstrong's new formulation (p. 13) of the challenge which he identifies with the Problem of Universals:

What is it for a class to mark off a type?

Instead of saying that a class 'marks of a type' we can say that that class is '*natural*'.

6 A possible response for the Class Nominalist

What if the Class Nominalist answered Armstrong's challenges using the same technique that was used to answer the question 'What is it to be an electron?':

- (13) For x and y to resemble one another is for the ordered pair $\langle x, y \rangle$ to be a member of the class of ordered pairs whose members resemble one another.
- (14) For a class to be natural is for it to be a member of the class of natural classes.